

CS-UY 2214 — Recitation 10

Introduction

Complete the following exercises. Unless otherwise specified, put your answers in a plain text file named `recitation10.txt`. Number your solution to each question. When you finish, submit your file on Gradescope. Then, in order to receive credit, you must ask your TA to check your work. Your work should be completed and checked during the recitation session.

Problems

1. As we discussed in class, an address can be divided into parts representing the tag, index, and block offset.

Below, we show how a 16-bit address can be divided for use in a particular direct-mapped cache. Each field is expressed as a range of bits (similar to how we express bit ranges in Verilog).

Tag	Index	Offset
15:10	9:4	3:0

For these questions, assume each memory cell contains 1 byte.

- (a) What is the blocksize of this cache? Give your answer in bytes.
 - (b) What is the size of the entire cache (excluding metadata)? Give your answer in bytes.
2. Assume a direct-mapped cache with 8 rows and a blocksize of 1 cell. The following memory addresses are accessed in sequence. For each of the following memory addresses, indicate (a) its row, (b) its tag, and (c) whether it will result in a hit, a miss, or a miss with eviction.
[12, 13, 6, 0, 2, 5, 14, 14, 1, 8]
 3. Repeat the previous exercise, but this time assume a direct-mapped cache with 4 rows and a blocksize of 2 cells.
 4. Repeat the previous exercise, but this time assume a direct-mapped cache with 2 rows and a blocksize of 4 cells.
 5. Assume a system with 32-bit pointers and 512-byte blocks. Its cache is 2 MB, excluding metadata. Each memory cell is one byte.
 - (a) How many blocks can be stored in the cache?
 - (b) How many bits is the block offset?
 - (c) Assuming a direct-mapped cache, how many bits is the index (i.e. the component of the address that indicates the cache row)?
 - (d) Assuming a direct-mapped cache, how many bits is the tag?

